



SET ~ 2

Series : PIZY70

कोड नं.
Q.P. Code

56/3/2

रोल नं.

Roll No.

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परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

नोट / NOTE

- (I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 6 हैं।
Please check that this question paper contains 6 printed pages.
- (II) कृपया जाँच कर लें कि इस प्रश्न-पत्र में 18 प्रश्न हैं।
Please check that this question paper contains 18 questions.
- (III) प्रश्न-पत्र में दाहिने हाथ की एयर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख्य-पृष्ठ पर लिखें।
Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- (IV) कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में यथा स्थान पर प्रश्न का क्रमांक अवश्य लिखें।
Please write down the serial number of the question in the answer-book at the given place before attempting it.
- (V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक परीक्षार्थी केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period.



रसायन विज्ञान (सैद्धांतिक)

CHEMISTRY (Theory)



निर्धारित समय: 2 घंटे

Time Allowed : 2 hours

अधिकतम अंक : 50

Maximum Marks : 50

56/3/2*



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P.T.O.

General Instructions:

Read the following instructions very carefully and strictly follow them :

- (i) This question paper contains **18** questions. **All** questions are **compulsory**.
- (ii) This Question paper is divided into **FIVE** sections- Sections **A, B, C, D** and **E**.
- (iii) In **Section A** : Question numbers **1** to **5** are Multiple Choice (MCQ) type questions. Each question carries **1** mark.
- (iv) In **Section B**: Question numbers **6** to **9** are Very Short Answer (VSA) type questions. Each question carries **2** marks.
- (v) In **Section C**: Question numbers **10** to **13** are Short Answer (SA) type questions. Each question carries **3** marks.
- (vi) In **Section D** : Question number **14** and **15** are Case Study-Based questions carrying **4** marks each.
- (vii) In **Section E** : Question numbers **16** to **18** are Long Answer (LA) type questions. Each question carries **5** marks.
- (viii) There is no overall choice given in the question paper. However, an internal choice has been provided in few questions in all the Sections except Section A.
- (ix) Kindly note that there is a separate question paper for Visually Impaired candidates.
- (x) Use of calculators is **NOT** allowed.

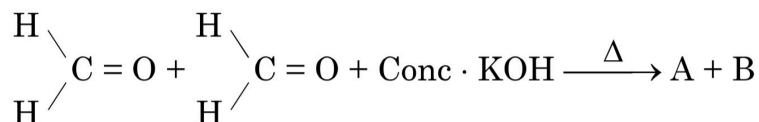
SECTION - A

This section contains 5 Multiple Choice Questions carrying 1 mark each.

1. The molar ionic conductivity of Al^{3+} and SO_4^{2-} are $189 \text{ S cm}^2 \text{ mol}^{-1}$ and $160 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. The value of limiting molar conductivity of $\text{Al}_2(\text{SO}_4)_3$ will be : **1**
 - (A) $198 \text{ S cm}^2 \text{ mol}^{-1}$
 - (B) $858 \text{ S cm}^2 \text{ mol}^{-1}$
 - (C) $588 \text{ S cm}^2 \text{ mol}^{-1}$
 - (D) $891 \text{ S cm}^2 \text{ mol}^{-1}$
2. A 1% solution of solute 'X' is isotonic with a 6% solution of sucrose (molar mass = 342 g/mol). The molar mass of solute 'X' is : **1**
 - (A) 34.2 g/mol
 - (B) 57 g/mol
 - (C) 114 g/mol
 - (D) 3.42 g/mol



3. Consider the following reaction :



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Identify A and B from the given options :

- (A) A — Methanol, B — Potassium formate
 (B) A — Ethanol, B — Potassium formate
 (C) A — Methanal, B — Ethanol
 (D) A — Methanol, B — Potassium acetate
4. Match the reagents required for the following reactions :
- I. Oxidation of primary alcohols to aldehydes (p) NaBH_4
 II. Butan-2-one to Butan-2-ol (q) 85% phosphoric acid at 440 K
 III. Bromination of Phenol to 2, 4, 6-Tribromophenol (r) PCC
 IV. Dehydration of propan-2-ol to propene (s) Bromine water
- (A) I — (r), (II) — (p), (III) — (s), (IV) — (q)
 (B) I — (q), (II) — (r), (III) — (p), (IV) — (s)
 (C) I — (s), (II) — (q), (III) — (p), (IV) — (r)
 (D) I — (p), (II) — (s), (III) — (r), (IV) — (q)
5. Nucleophilic addition to Grignard reagents to ketones followed by hydrolysis with dilute acid forms :
- (A) Alkene (B) Primary alcohol
 (C) Tertiary alcohol (D) Secondary alcohol

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SECTION - B

This section contains 4 Very Short Answer (VSA) type questions carrying 2 marks each.

6. (a) Would you expect benzaldehyde to be more reactive or less reactive in nucleophilic addition reactions than propanal? Explain your answer.

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- (b) Arrange the following compounds in increasing order of their reactivity in nucleophilic addition reactions.
- Ethanal, propanal, Propanone, Butanone
 - Benzaldehyde, p-Tolualdehyde, p-Nitrobenzaldehyde, Acetophenone.
7. A solution containing 60 g of a non-volatile solute in 250 g of water freezes at 270.67 K. Calculate the molar mass of the solute. 2
(K_f of water = $1.86 \text{ K kg mol}^{-1}$).
8. (a) Which of the following compounds will react more rapidly by $\text{S}_\text{N}1$ reaction and why? 2
- $$\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{Br} \text{ or } \text{CH}_3\text{CH}_2\text{Br} \\ | \\ \text{CH}_3 \end{array}$$
- (b) Arrange the following compounds in the increasing order of their boiling points :
Bromoform, Dibromoethane, Chloromethane, Bromoethane.
9. How will you convert the following : 2
- Toluene to benzoic acid.
 - Ethanol to 3-hydroxybutanal

SECTION - C

This section contains 4 Short Answer (SA) type questions carrying 3 marks each.

10. Write chemical equations for the following reactions : (Do any **three**) 3
- Hydroboration — oxidation reaction
 - Williamson Synthesis
 - Friedel-Crafts Alkylation of Anisole
 - Reimer-Tiemann Reaction
11. (a) Give chemical tests to distinguish between the following pairs of compounds : 3
- Phenol and benzoic acid
 - Propanal and propanone
- (b) Which one of the following compounds is a stronger acid and why?
 $\text{CH}_2\text{FCH}_2\text{CH}_2\text{COOH}$ or $\text{CH}_3\text{CHFCH}_2\text{COOH}$



12. (a) Write the IUPAC name of the given compound : 3
 $\text{CH}_2 = \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{Cl}$
- (b) Why is thionyl chloride preferred for preparing alkyl halides from alcohols?
- (c) What happens when methyl bromide reacts with KCN ?
13. Calculate emf of the following cell at 298 K : 3
 $\text{Sn}/\text{Sn}^{2+} (0.001 \text{ M}) \parallel \text{H}^+ (0.01 \text{ M}) \mid \text{H}_{2(\text{g})} (1 \text{ bar}) \mid \text{Pt}_{(\text{s})}$
 Given : $E^\circ (\text{Sn}^{2+}/\text{Sn}) = -0.14 \text{ V}$, $E^\circ (\text{H}^+/\text{H}_2) = 0 \text{ V}$ ($\log 10 = 1$)

SECTION - D

This section consists of 2 case-based/source-based questions carrying 4 marks each.

14. In a galvanic cell, chemical energy of a redox reaction is converted into electrical energy whereas in an electrolytic cell the redox reaction occurs on passing electricity. The simplest galvanic cell is in which Zn rod is placed in a solution of ZnSO_4 and Cu rod is placed in a solution of CuSO_4 . The two rods are connected by a metallic wire through the voltmeter. The two solutions are joined by a salt bridge. The difference between the two electrode potentials of the two electrodes is known as electromotive force. In the process of electrolysis, the decomposition of a substance takes place by passing an electric current. One mole of electric charge when passed through a cell will discharge half a mole of a divalent metal ion such as Cu^{2+} . This was first formulated by Faraday in the form of laws of electrolysis. 4

Answer the following questions :

- (a) What is the function of a salt bridge in a galvanic cell ?
- (b) When does galvanic cell behave like an electrolytic cell ?
- (c) Can copper sulphate solution be stored in a pot made of zinc ? Explain with the help of the value of E° cell.
 $(E^\circ \text{ Cu}^{2+} / \text{Cu} = 0.34 \text{ V})$
 $(E^\circ \text{ Zn}^{2+} / \text{Zn} = -0.76 \text{ V})$
- (d) How much charge in terms of Faraday is required for the following :
 (i) 1 mol of MnO_4^{2-} to Mn^{2+}
 (ii) 1 mol of H_2O to O_2



15. Phenols undergo electrophilic substitution reactions readily due to the strong activating effect of OH group attached to the benzene ring. Since, the OH group increases the electron density more to o— and p— positions therefore OH group is ortho, para directing. Reimer-Tiemann reaction is one of the examples of aldehyde group being introduced on the aromatic ring of phenol, ortho to the hydroxyl group. This is a general method for the ortho-formulation of phenol. 4

Answer the following questions :

- (a) What happens when phenol reacts with (i) Br_2/CS_2 , (ii) Conc. HNO_3
- (b) Why phenol does not undergo protonation readily ?
- (c) Which is a stronger acid- phenol or cresol? Give reason.
- (d) Write the IUPAC name of the product formed in the Reimer-Tiemann reaction.

SECTION - E

This section consists of 3 Long Answer (LA) type questions carrying 5 marks each.

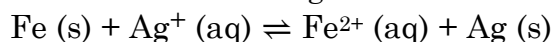
16. An organic compound (A) with molecular formula $\text{C}_8\text{H}_8\text{O}$ forms an orange red precipitate with 2, 4-DNP reagent and gives yellow precipitate on heating with iodine in the presence of sodium hydroxide. It neither reduces Tollens' nor Fehling's reagent, nor does it decolourise Bromine water in Baeyer's reagent. On drastic oxidation with chromic acid, it gives Carboxylic acid (B) having molecular formula $\text{C}_7\text{H}_6\text{O}_2$. Identify the compounds (A) and (B) and explain the reactions involved. 5

17. (a) Predict whether Van't Hoff factor will be less or greater than one, when ethanoic acid is dissolved in benzene. 5

(b) Define ideal solution.

- (c) Calculate the mass of CaCl_2 (molar mass = 111 g/mol) to be dissolved in 500 g of water to lower freezing point by 2 K, assuming that CaCl_2 undergoes complete dissociation (K_f for water = 1.86 K kg/mol)

18. (a) Calculate the ΔG and $\log K_c$ for the following cell reaction :



Given : $E^\circ (\text{Fe}^{2+} / \text{Fe}) = 0.44 \text{ V}$, $E^\circ (\text{Ag}^+ / \text{Ag}) = + 0.8 \text{ V}$

1 F = 96500 C/mol

- (b) Write any two advantages of the fuel cells over primary and secondary batteries. 5

